CI/CD and Jenkins

React Application with Express Server

Ido Carmi & Yuval Arviv

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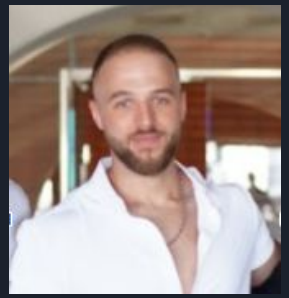
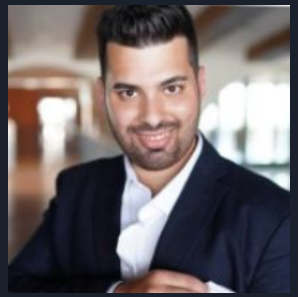


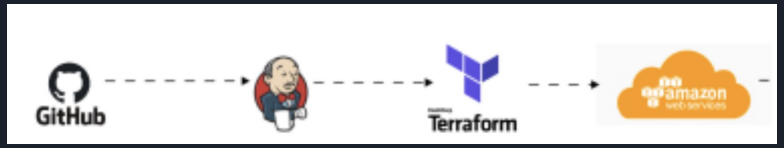
Ido Carmi
Pelephone - Network Operation Center
25 Years old, live in Ramat Gan



Yuval Arviv
Pelephone - Network Operation Center
23 Years old, live in Mazkeret batya

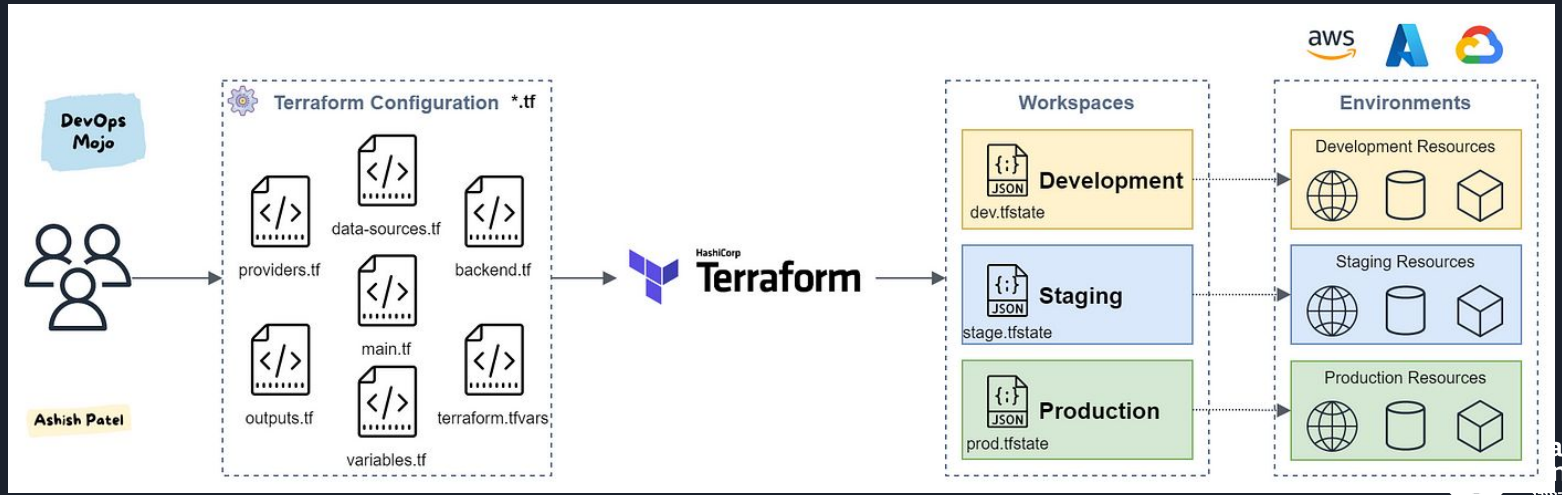
What is Jenkins

- **Automation Server**: Jenkins is an automation server that streamlines software development processes.
- **Extensible Plugins**: It's highly customizable and extends functionality through plugins.
- **Pipeline as Code**: Jenkins supports defining workflows as code, enhancing automation.
- **User-Friendly Interface**: Offers a web-based interface for easy job configuration.



What is Terraform

- **Infrastructure as Code**: Terraform is a tool used for provisioning and managing cloud infrastructure.
- **Multi-Cloud Support**: Terraform supports multiple cloud providers (e.g., AWS, Azure, Google Cloud) and other infrastructure platforms.
- **Plan and Apply**: Terraform provides a "plan" feature to preview changes before applying them, enhancing safety and predictability in infrastructure changes.



Project Overview

Goal: Create a seamless CI/CD pipeline for a React application with an Express server.

Setup: The project was initiated with Create React App, and an Express server was integrated in the server directory. The proxy key in package.json handles server proxying.

Objective: Achieve automated building, testing, and deployment processes to ensure efficient development and delivery.

Running the Application

- Clone the project
- Install dependencies
- Application runs on port 3000

```
yuval@yuval:~/Desktop/project/red-project$ ls
Jenkinsfile  README.md  frontend  package-lock.json  server  terraform  test
yuval@yuval:~/Desktop/project/red-project$
```

```
# Make port 3000 available to the world outside this container
EXPOSE 3000
```

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NPM Install / Start

- Run 'npm install'
- Run 'npm start'

```
yuval@yuval:~/Desktop/red-project/server$ npm install

up to date, audited 177 packages in 2s

14 packages are looking for funding
  run `npm fund` for details
```

```
yuval@yuval:~/Desktop/project/red-project/server$ npm start

> react-express-starter@0.1.0 start
> node index.js

Express server is running on localhost:3000
```

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Running Tests

- Change directory to test folder
- Install requirements with 'pip3 install -r requirements.txt'
- Run tests using 'pytest test/test.py'

```
yuval@yuval:~/Desktop/red-project$ pytest test/test.py
============================================================ test session starts ============================================================
platform linux -- Python 3.10.12, pytest-7.2.1, pluggy-1.0.0
rootdir: /home/yuval/Desktop/red-project
collected 2 items

test/test.py ..                                                                                                                      [100%]

============================================================= 2 passed in 0.00s =============================================================
```

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Building Docker Images

- Build two Docker files (server and frontend)

```
frontend                    frontend        49d7db6cb063
```

```
server                      server          53df9c82936e
```

```
# Use an official Node runtime as the parent image
FROM node:14

# Set the working directory in the container to /app
WORKDIR /app

# Copy package.json and package-lock.json to the workdir
COPY package*.json ./

# Install any needed packages specified in package.json
RUN npm install

# Bundle app source inside the docker image
COPY . .

# Make port 3000 available to the world outside this container
EXPOSE 3001

# Run server.js when the container launches
CMD ["npm", "start"]
```

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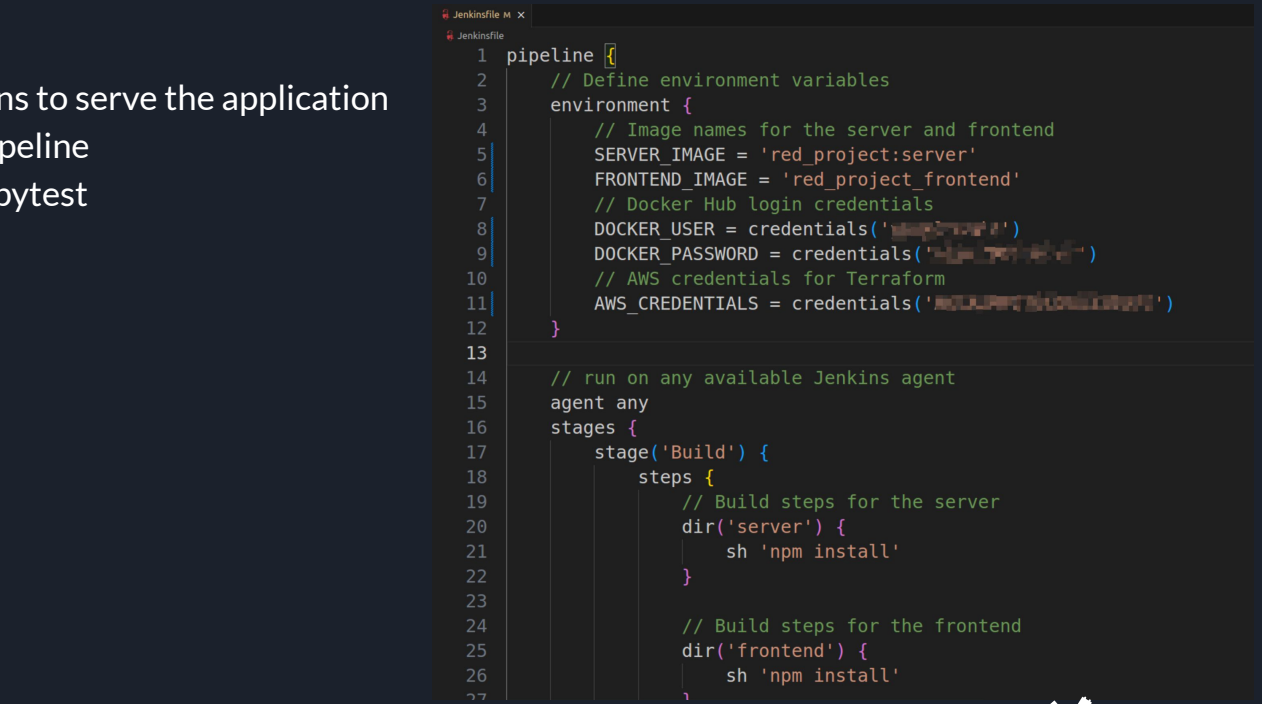
Setting up Jenkins CI Pipeline

- Configure Jenkins to serve the application
- Implement CI pipeline
- Run tests using pytest



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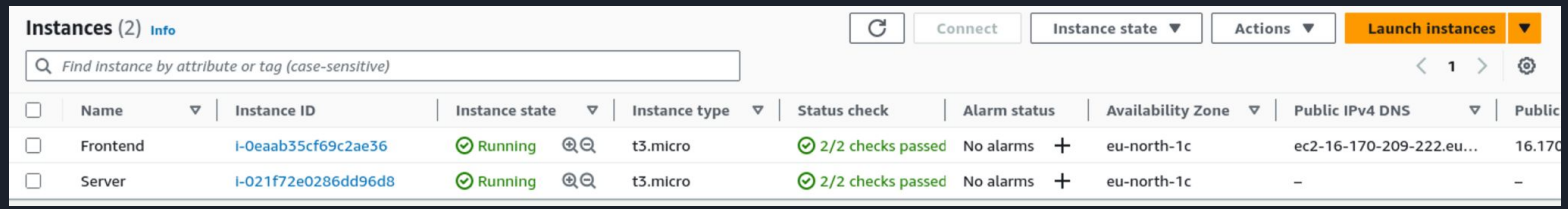
Deploying with Terraform

- Use Jenkins to build Terraform CD
- Deploy two EC2 instances on AWS

```
yuval@yuval:~/Desktop/red-project/terraform$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
  + create

Terraform will perform the following actions:
```



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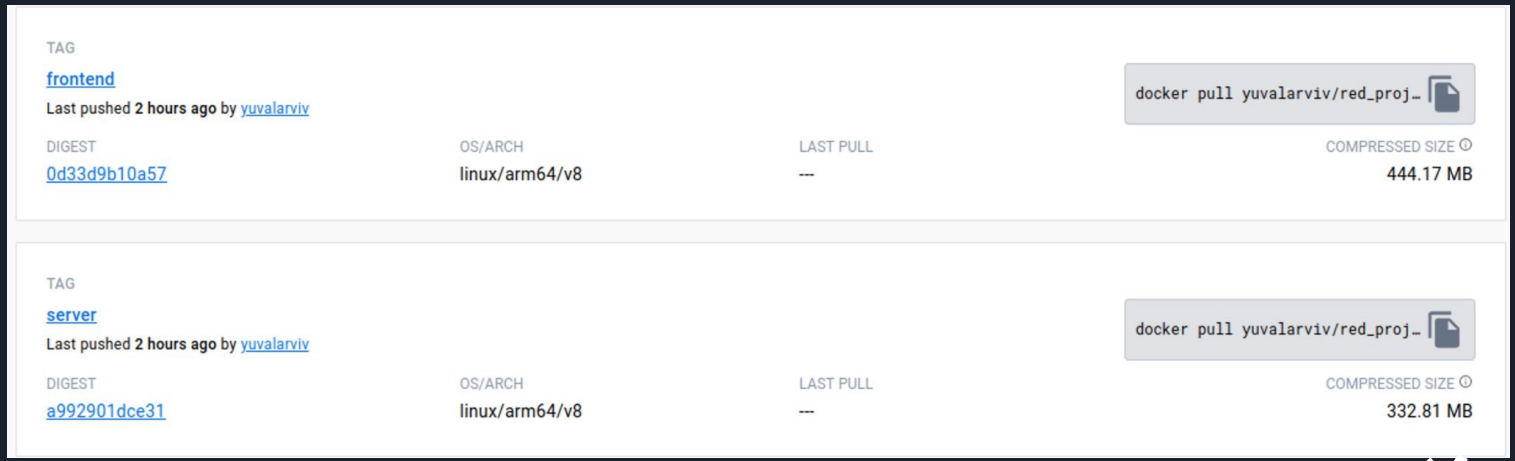
Deploying Docker Images

- Deploy the first Docker image to the first application
- Deploy the second Docker image to the second application



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Conclusion

- The CI/CD pipeline automates the application deployment
- Jenkins, Docker, and Terraform streamline the process
- Efficiently manage and deploy your React-Express application.

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Thank you

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